

Sepideh Maleki

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Work Authorization: U.S. Permanent Resident (Green Card)

Selected Projects

Mechanism-aware Perturbation Transformer

Postdoctoral Researcher, Genentech

- Developed a counterfactual gene-token transformer for drug–cell perturbation prediction using single-cell transcriptomics, improving retrieval performance by up to +40% MRR and gene-level prediction accuracy by up to +30% Pearson compare to SOTA.
- Designed a sparse mixture-of-experts mechanism routing module that learned interpretable latent biological programs (e.g., stress response, ECM remodeling, lineage identity) directly from perturbation data.
- In review

LLM-Guided Retrieval for Molecular Perturbation Prediction

- Built a retrieval-based system using LLM reasoning to predict single-cell transcriptional responses to unseen drug–cell combinations.
- Replaced end-to-end regression with LLM-driven neighbor selection + aggregation, improving generalization to unseen cell types and boosting sign accuracy by >20% in open-world evaluation.
- Benchmarked against PCA+Ridge, chemical kNN, drug-mean, and cell-mean baselines on large-scale single-cell perturbation datasets.
- In review

Multi-modal Foundation Models for Predicting Molecular Perturbation

Postdoctoral Researcher, Genentech

- Led a project to fine-tune single-cell foundation models (FMs) pre-trained on tens of millions of single cells to predict molecular perturbations.
- Introduced a drug-conditional adapter, enabling efficient fine-tuning by training less than 1% of the original FM, which preserved the rich biological representation from pre-training.
- Developed a strategy that facilitates not only the prediction of cellular responses to novel drugs but also zero-shot generalization to unseen cell types.
- Spotlight paper at MLGenX'25

Score-based Explainability for Graph Representations

Postdoctoral Researcher, Genentech

- Collaborated on the development of a novel framework, grXAI, aimed at enhancing graph representation explainability.
- grXAI generalizes existing score-based graph explainers to identify subgraphs most responsible for constructing the latent representation of input graphs.
- Implemented as a wrapper around existing methods, grXAI allows for the explanation of graph representations through connected subgraphs, improving human interpretability.
- Published in TMLR'24 and adopted by multiple internal teams.

Representation Learning on Hypergraphs

PhD Researcher, UT Austin

- Led a project focused on scaling machine learning techniques for large (hyper)graphs.
- Developed a hierarchical hypergraph embedding system capable of generating embeddings for hypergraphs with millions of nodes and hyperedges in just minutes, compared to the days required by existing methods.
- Demonstrated the efficacy of these embeddings in classification and prediction tasks on hypergraphs.
- Published in Network Science 2022.

Deterministic Hypergraph Partitioning

PhD Researcher, UT Austin

- Designed and implemented the first deterministic and parallel hypergraph partitioner, surpassing state-of-the-art partitioners in both running time and partition quality.
- Achieved deterministic partitions, a significant advancement over existing methods, ensuring consistent and reliable results.
- Published in PPOPP'21, and currently utilized in a DARPA ERI project on asynchronous circuit design led by Rajit Manohar at Yale.
- Adopted by other teams in the DARPA ERI program, including Pierre-Emmanuel Gaillardon's group at the University of Utah for logic synthesis.

Education

PhD, the University of Texas at Austin, August 2022

Master of Science in Computer Science, Texas State University, May 2016

Bachelor of Science in Computer Science, Kharazmi University, Iran, June 2011

Work Experience

Postdoctoral Associate – Genentech (2022 - Present)

Graduate Research Assistant – University of Texas at Austin (2016 - 2022)

Advised by Professor Keshav Pingali

Biomedical Data Science– Stanford University (Summer 2020)

Advised by Professor Dennis P Wall

Software Engineering Intern – Google, Inc. (Summer 2017)

Developed a benchmark that is representative of the kind of code that Google Executes

Researcher – Texas State University (Summer 2016)

Graduate Research Assistant – Texas State University (2015, Spring 2016)

Advised by Professor Martin Burtcher

Qualifications

Programming Languages

Python, C/C++, Java, CUDA, C#.NET, R, Matlab

Libraries & Frameworks

PyTorch, TensorFlow, LangGraph, Scikit-learn, Weights & Biases (W&B), NumPy, Pandas, Matplotlib, Hugging Face Transformers

Machine Learning & AI Expertise

Supervised & Unsupervised Learning, Foundation Models, Large Language Models (LLMs), Graph Neural Networks (GNNs), Model Explainability & Interpretability, Transfer Learning, Hyperparameter Optimization

Scientific Computing & Data Engineering

Apache Arrow, Dask, HDF5, SQL, Scanpy, AnnData, Seurat

DevOps & Tools

Docker, Git, Linux, Jupyter, VS Code, Slurm, REST APIs, CI/CD (GitHub Actions)

Cloud & Platforms

AWS, Google Cloud, Azure, Hugging Face Hub

Honors, Awards, and Activities

Panelist at the Learning Meaningful Representations of Life (LMRL) workshop at ICLR 2025

ICLR'24 workshop organizer – 1st Machine Learning for Genomics Exploration Workshop (MLGenX) ICLR 2024

Graduate School fellowship – The Graduate School at the U. of Texas at Austin (2022)

AMS Mathematics Research on Models and Methods for Sparse (Hyper)Networks – funded participant (2022)

Spring school – Lake Como School of Advanced Studies; Complex Networks (2022)

PSB session organizer – Graph Representations and Algorithms in Biomedicine (2022)

PPOPP artifact evaluation committee – (2021)

Best Workshop paper – WOSET (2019)

First Place at the Conference of Southern Graduate Schools (CSGS) competition (2017)

Outstanding Master's Student Award – College of Science and Engineering at Texas State U. (2016)

Computer Science Endowed Scholarship – Department of Computer Science at Texas State University (2016)

Graduate Research Excellence Award – Department of Computer Science at Texas State University (2015)

Publications and Workshops

1. Russell Littman*, Jacob Levine*, **Sepideh Maleki**, Yongju Lee, Vladimir Ermakov, Lin Qui, Alexander Wu, Kexin Huang, Romain Lopez, Gabriele Scalia, Tommaso Biancalani, David Richmond, Aviv Regev&, Jan-Christian Huetter, Gene-embedding-based prediction and functional evaluation of perturbation expression responses with PRESAGE. *biorxiv* 2025.
2. **Sepideh Maleki**, Jan-Christian Huetter, Kangway V. Chuang, Gabriele Scalia, q, Efficient Fine-Tuning of Single-Cell Foundation Models Enables Zero-Shot Molecular Perturbation Prediction, *MLGenX 2025* (spotlight paper)
3. Ehsan Hajiramezanali, **Sepideh Maleki**, Max W Shen, Kangway V. Chuang, Tommaso Biancalani, Gabriele Scalia. Score-based Explainability for Graph Representations, *TMLR* 2024
4. **Sepideh Maleki**, Jan-Christian Huetter, Kangway V. Chuang, Gabriele Scalia, Tommaso Biancalani, Leveraging Single-Cell Foundation Models for Molecular Perturbation Prediction, *MLCB* 2024
5. Jürgen Kraitschgau, Daniel Kaiser, Oliver Alvarado Rodriguez, Ilya Amburg, Jessalyn Bolkema, Thomas Grubb, Fangfei Lan, **Sepideh Maleki**, Phil Chodrow, Bill Kay. Community detection in hypergraphs via mutual information maximization. *Nature Scientific Report* 2024.
6. **Sepideh Maleki**, Kangway V. Chuang, Gabriele Scalia, Tommaso Biancalani, . Learning to Explain Hypergraph Neural Networks, *Neurips 2023 workshop*.
7. **Sepideh Maleki**, Josh Vekhter, Keshav Pingali. Predicting and Classifying Hyperedges through Transfer Learning, *arxiv* 2022.
8. **Sepideh Maleki**, Dennis P Wall, Keshav Pingali, HyperNetVec: Fast and Scalable Hierarchical Embedding for Hypergraphs, *NetSci-X* 2022.
9. Indels in SARS-CoV-2 occur at template-switching hotspots, Brianna Chrisman, Kelley Paskov, Nate Stockham, Kevin Tabatabaei, Jae-Yoon Jung, Peter Washington, Maya Varma, Min Woo Sun, **Sepideh Maleki**, Dennis Paul Wall, *BioData Mining* 2021
10. **Sepideh Maleki**, Udit Agarwal, Martin Burtscher, and Keshav Pingali, BiPart: A Parallel and Deterministic Hypergraph Partitioner. *ACM SIGPLAN Annual Symposium Principles and Practice of Parallel Programming(PPoPP)* 2021.
11. **Sepideh Maleki**, Udit Agarwal, and Keshav Pingali, Hypergraph Partitioning via Geometric Embeddings. *3rd Workshop on Open-Source EDA Technology (WOSET)*, November 5, 2020.
12. Yi-Shan Lu, Samira Ataei, Jiayuan He, Wenmian Hua, **Sepideh Maleki**, Yihang Yang, Keshav Pingali and Rajit Manohar. An Open-Source EDA flow for Asynchronous Logic. *IEEE Design & Test* 2020.
13. Samira Ataei, Jiayuan He, Wenmian Hua, Yi-Shan Lu, **Sepideh Maleki**, Yihang Yang, Keshav Pingali, Rajit Manohar. Toward a Digital Flow for Asynchronous VLSI Systems. In *2nd Workshop on Open-Source EDA Technology (WOSET)*, Westminster, CO, November 7, 2019. **Second best place** WOSET best EDA tool award.
14. Yi-Shan Lu, Samira Ataei, Jiayuan He, Wenmian Hua, **Sepideh Maleki**, Yihang Yang, Martin Burtscher, Keshav Pingali and Rajit Manohar. Parallel Tools for Asynchronous VLSI Systems. *Workshop on Open-Source EDA Technology. Co-located with ICCAD 2018. WOSET'18*
15. Yi-Shan Lu, Samira Ataei, Jiayuan He, Wenmian Hua, **Sepideh Maleki**, Yihang Yang, Martin Burtscher, Keshav Pingali, Rajit Manohar. Parallel Tools for Asynchronous VLSI Systems. In *1st Workshop on Open-Source EDA Technology (WOSET)*, San Diego, CA, November 8, 2018
16. **S. Maleki**, and M. Burtscher. "Automatic Hierarchical Parallelization of Linear Recurrences", *ASPLOS'18*
17. **S. Maleki**, A. Yang, and M. Burtscher. "Higher-Order and Tuple-Based Massively-Parallel Prefix Sums." *ACM SIGPLAN Conference on Programming Language Design and Implementation. June 2016. PLDI'16*

18. **S. Maleki.** “Generalizing Massively-Parallel Prefix-Sum Computations to Multiple Dimensions and Higher Orders.” *Programming Languages Colloquium at the University of Texas at Austin*. April 2016
19. **S. Maleki.** “A New High-Speed Prefix-Scan Algorithm for GPUs.” *GPU Technology Conference*. April 2016.